UNIT 1 · 9 hours

Phasor Representation of Alternating Quantities



Circuit Fundamentals

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

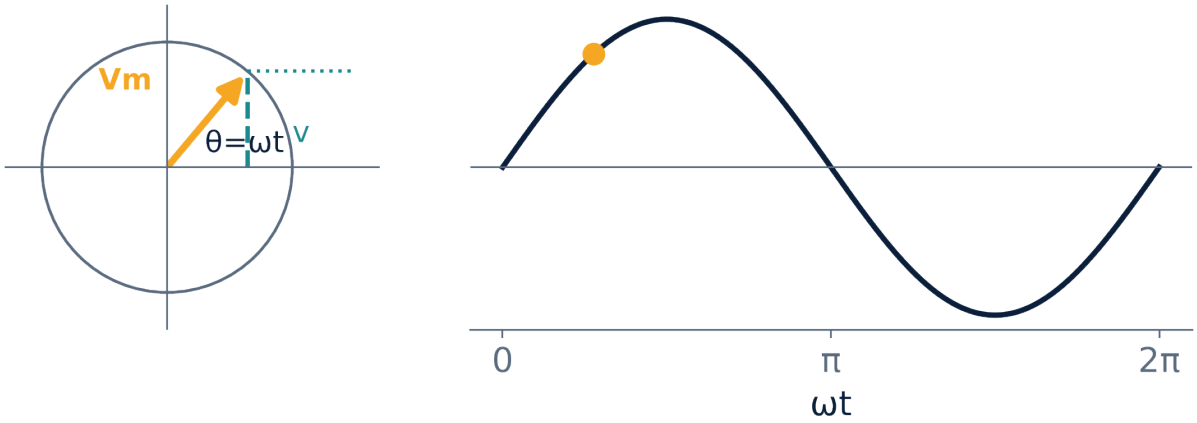
- Represent a sinusoid as a rotating phasor
- Define phase difference, lead and lag
- Add sinusoids using phasors
- Use the polar and rectangular forms

Syllabus context

Basic nodal and mesh analysis excited by independent DC voltage sources, Power and Energy. Generation of sinusoidal voltage, frequency, average value, RMS value, form factor and peak factor, phasor representation of alternating quantities.

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Rotating phasor and its projection



- Phasor OA of length Vm rotates anticlockwise at ω = 2πf.
- Vertical projection = instantaneous value v = Vm sin ωt.
- One revolution = 2π rad = one cycle.
- All same-frequency phasors rotate together, so only their lengths and relative angles matter.

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Phase difference

In phase

Two waves reach zero and peak together. Phase difference 0°.

Leading

Wave that reaches its peak earlier is leading, e.g. $i = I_m \sin(\omega t + \varphi)$ leads v.

Lagging

Wave that reaches its peak later is lagging, e.g. $i = I_m \sin(\omega t - \varphi)$.

In phasor diagrams, anticlockwise from the reference is leading.

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Forms of a phasor

Form	Expression	Use
Polar	$V\angle\theta$	multiply / divide
Rectangular	$a + jb$	add / subtract
Conversion	$a = V\cos\theta,\ b = V\sin\theta$	$V = \sqrt{(a^2 + b^2)},\ \theta = \tan^{-1}(b/a)$
Operator j	$j = 90°$ rotation, $j^2 = -1$	$I\angle 90° = jI$

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Example: adding two voltages

Problem $v_1 = 10 \sin \omega t$ and $v_2 = 5 \sin(\omega t + 90°)$. Find $v_1 + v_2$ (peak values).

- Phasors: $V_1 = 10\angle 0° = 10 + j0$, $V_2 = 5\angle 90° = 0 + j5$
- Sum $= 10 + j5$
- $|V| = \sqrt{(10^2 + 5^2)} = 11.18$, $\theta = \tan^{-1}(5/10) = 26.57°$

Answer: $v = 11.18 \sin(\omega t + 26.57°)$

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Why phasors are useful

- Reduce differential equations to algebra.
- Voltages in a series circuit add as phasors, never arithmetically.
- Phasor diagrams reveal lead/lag and power factor at a glance.
- RMS values are normally used as phasor lengths.

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Key takeaways

- A phasor rotates at ω; its vertical projection is the instantaneous value
- Phase difference = angle between phasors
- Add in rectangular, multiply in polar form
- Series circuit voltages add as phasors

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Practice questions

Q1. Express v = 100 sin(ωt − 30°) as a phasor in polar and rectangular form.

Q2. Add i1 = 4 sin ωt and i2 = 3 sin(ωt − 90°).

Q3. Which leads: v = sin(ωt + 20°) or i = sin(ωt − 40°)? By how much?

Q4. Why must all phasors in one diagram have the same frequency?